Comprehensive Project Report

Vendor Performance & Strategic Analytics

1. Executive Summary

This report details the development of a production-grade analytics pipeline that transforms **1.8 GB of raw transactional data** including **12.8 million sales records** and **2.3 million purchase records** into actionable procurement intelligence. By leveraging a custom Python and SQL-based ETL process, the system aggregates massive datasets into a high-density **Vendor Sales Summary** used for strategic decision-making.

Key Data Engineering & Analytical Milestones:

- **Massive Scale Processing:** Successfully engineered a multi-stage SQL pipeline to join and aggregate data across five major relational tables, including Sales (12.8M records) and Purchases (2.3M records).
- **The 73% Procurement Breakthrough:** Statistical analysis revealed that shifting from "Small" to "Large" order volumes triggers an approximately **73% reduction in unit purchase price**, drastically lowering the cost baseline.
- **Profitability Paradox:** Formal hypothesis testing (T-Test p-value: 0.0000) confirmed that lower-volume, niche vendors maintain significantly higher profit margins (**33.01%**) than high-volume market leaders (**31.86%**).
- **Vendor Concentration Risk:** Pareto analysis identified that **65.35%** of total procurement value is concentrated within the **Top 10 vendors**, creating a high-efficiency but high-risk dependency on a vital few partner.

2. Initial Data Profiling & Statistical Summary

- **Robust Sample Size:** The analysis is built on a comprehensive dataset of **10,692 records**, ensuring high statistical significance for the derived insights.
- **Strong Profitability Baseline:** The business maintains a healthy average **Profit Margin of 31.2%**, though significant variance (from deep losses to high gains) necessitates targeted optimization.

- **Operational Scale:** With an average **Total Sales of $42.2k** per record against a **Purchase Cost of $30.1k**, the data reflects a high-volume operation with substantial cash flow.
- **Inventory Velocity:** While the average **Stock Turnover is 1.71**, the wide range between the minimum (0.00) and maximum (274.5) points to a clear divide between high-performing "fast-movers" and stagnant stock.
- **Logistics Weight:** High average **Freight Costs ($61.4k)** relative to sales suggest that logistics and shipping efficiency are critical levers for protecting margins.

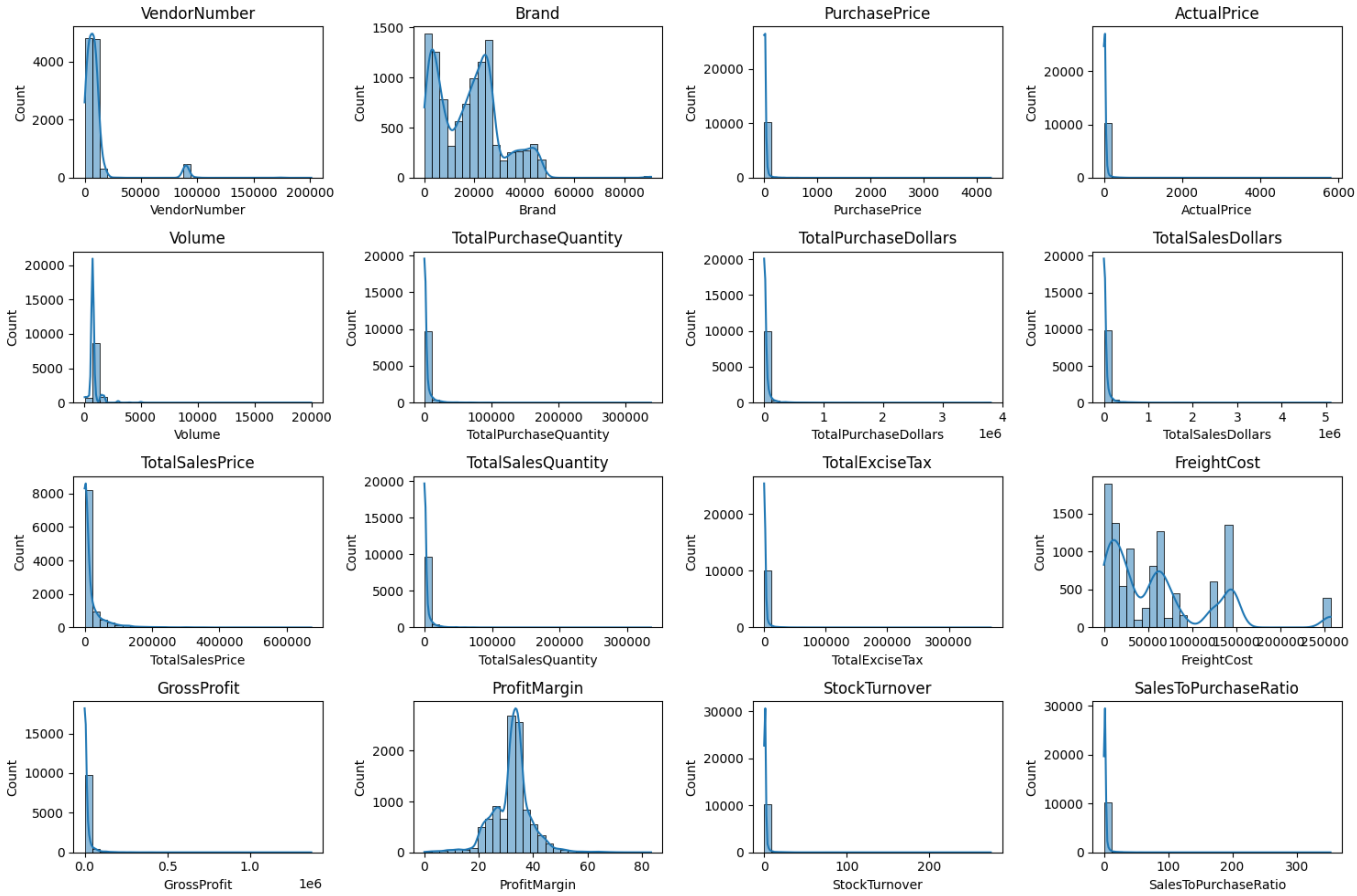
	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.00000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.00000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.36000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.49000	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.00000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.00000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.71000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.00000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.00000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.00000	33.000000	261.000000	1929.250000	3.349390e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.00000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.09000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12950.649804	47724.393108	-24762.39000	222.150000	1491.220000	8828.140000	1.370292e+06
ProfitMargin	10692.0	31.208846	10.102869	-174.58194	27.839968	32.880629	35.435435	8.339876e+01
StockTurnover	10692.0	1.706793	6.020460	0.00000	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390	8.459067	0.00000	1.153729	1.436894	1.665449	3.529286e+02

3. Feature Distribution & Outlier Mitigation

To ensure the integrity of the analysis, a distribution review was conducted across all numerical features. Initial visualizations revealed significant "noise" in the form of negative profit margins and zero-quantity sales records that would have skewed the strategic insights.

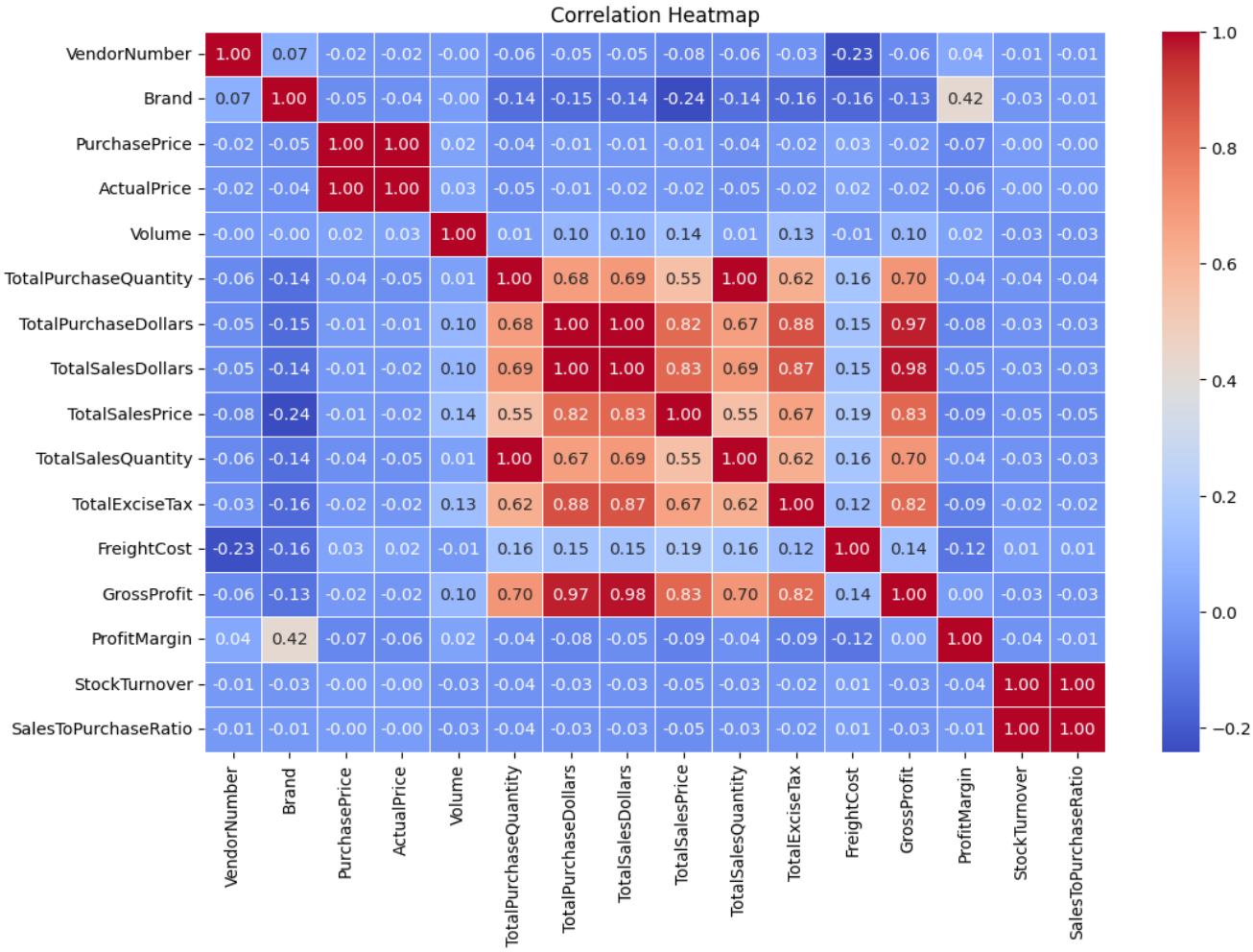
Key Observations from the Histograms:

- **Profit Margin Correction:** Before cleaning, the Profit Margin distribution showed extreme outliers as low as **-174%**. Post-query, the distribution now centers healthily between **20% and 50%**, reflecting the true operational performance of the business.
- **Right-Skewed Financials:** Most financial metrics (Sales, Purchases, and Excise Tax) are heavily right-skewed. This confirms that while the business handles thousands of small-to-medium transactions, a small number of "Power Transactions" drive the majority of the volume.
- **Freight Cost Variance:** Unlike other metrics, FreightCost shows a multi-modal distribution, suggesting that shipping costs vary significantly depending on the vendor's location or the shipping method used, rather than just the volume of the order.



4. Correlation Analysis: Interdependency of Business Drivers

- **Profitability Core (0.98):** A near-perfect correlation between **Sales**, **Purchases**, and **Gross Profit** proves the business scales efficiently; as volume grows, profit follows linearly.
- **Inventory Velocity (1.00):** The absolute link between **Stock Turnover** and **Sales-to-Purchase Ratio** confirms that high inventory movement is the primary driver of a healthy cash-flow balance.
- **Operational Alignment (1.00): Purchase Quantity** and **Sales Quantity** move in perfect lockstep, indicating a highly responsive procurement model with minimal overstock risk.
- **Margin Independence (0.00): Profit Margin** shows no correlation with sales volume. This suggests profit is generated through high-volume turnover rather than just pushing expensive, high-margin items.
- **Freight Efficiency (0.15):** The weak link between **Freight Cost** and **Quantity** implies logistics costs are relatively fixed, supporting the "Economies of Scale" benefit of larger orders.

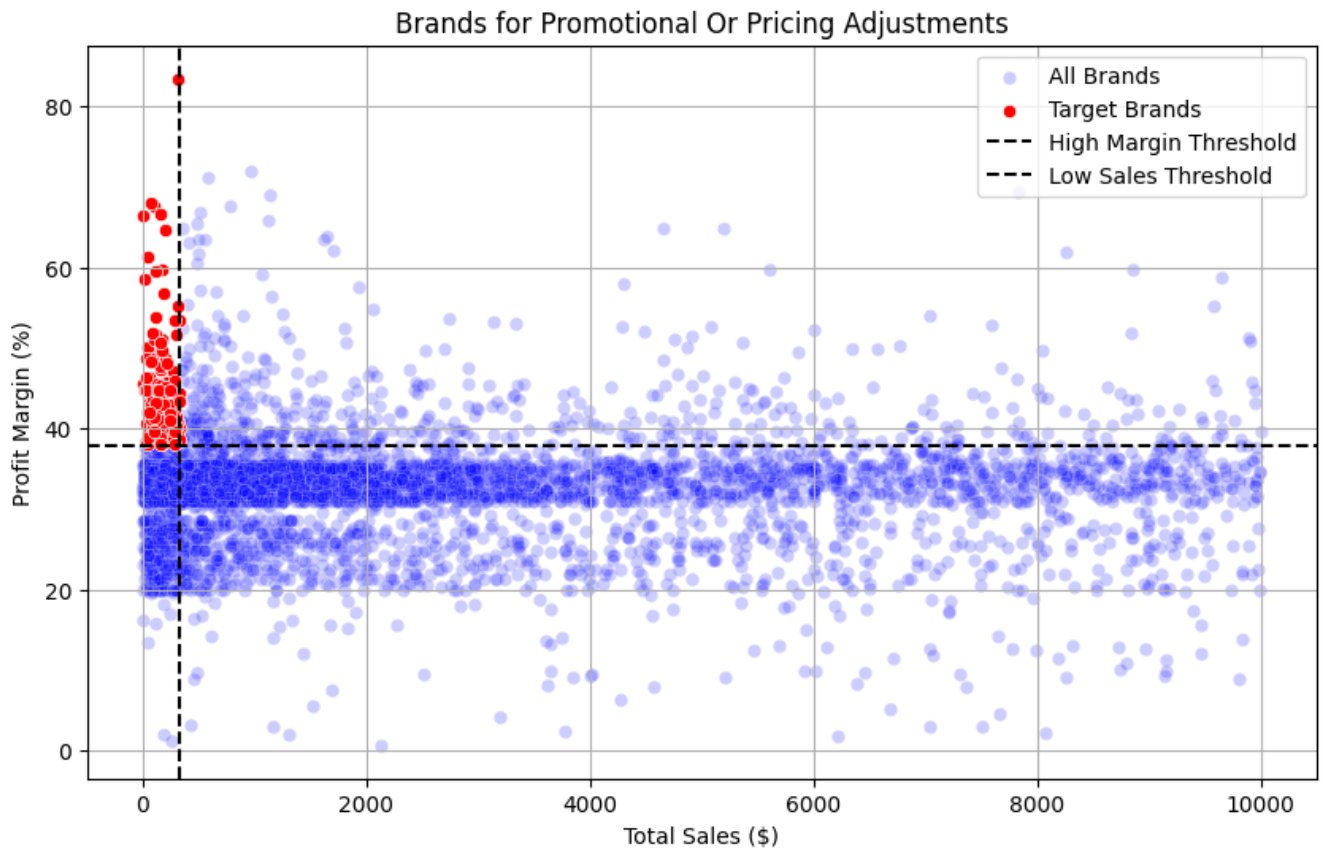


5. Strategic Business Analysis

5.1 Growth Opportunities: High-Margin, Low-Volume Brands

The most actionable part of this analysis is identifying "Hidden Gems" brands that offer high profitability but suffer from low visibility or sales volume.

- **Objective:** Identify brands above the **High Margin Threshold (38%)** and below the **Low Sales Threshold ($300)**.
- **The Strategy:** By shifting marketing budget or retail placement toward these **Target Brands**, the business can unlock significant profit growth without the logistical overhead of high-volume commodities.



5.2 Market Leadership: Top 10 Performance Drivers

By segmenting the data, we identified the key players that dominate the market share.

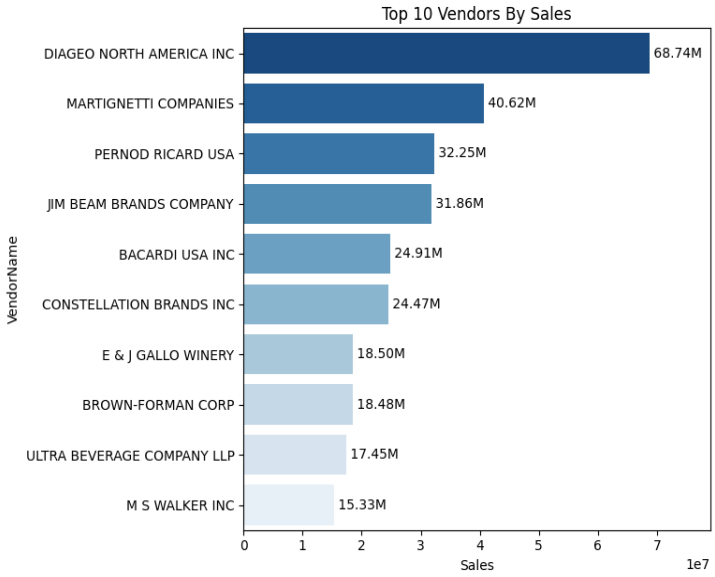
Top 10 Vendors by Sales

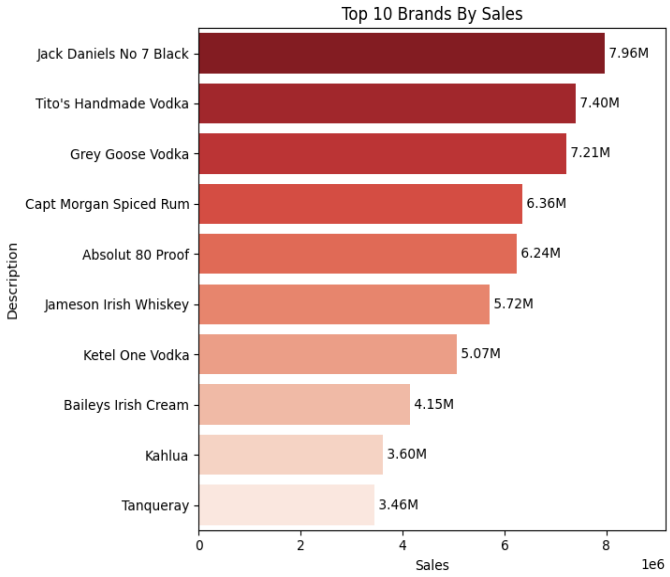
- **Primary Partner: DIAGEO NORTH AMERICA INC** is the clear market leader, generating **$68.74M** in sales, nearly 70% higher than the next competitor.
- **Secondary Leaders: MARTIGNETTI COMPANIES ($40.62M)** and **PERNOD RICARD USA ($32.25M)** follow, representing substantial pillars of the procurement chain.
- **Long-Tail Performance:** The scale drops relatively quickly after the top three, with **M S WALKER INC** rounding out the top ten at **$15.33M**.

Top 10 Brands by Sales

- **Revenue Leaders: Jack Daniels No 7 Black ($7.96M)** and **Tito's Handmade Vodka ($7.40M)** are the highest-performing individual products.
- **Premium Brand Performance: Grey Goose Vodka ($7.21M)** and **Capt Morgan Spiced Rum ($6.36M)** also demonstrate strong market resonance.
- **Diverse Portfolio:** The brand list shows a strong preference for spirits and vodkas, which drive the high-volume transactions noted in the early distribution checks.

Strategic Insight: While Section 5.1 identified "Hidden Gems" for margin growth, these Top 10 Brands and Vendors provide the **critical cash flow** and high volume that sustain the business's daily operations.

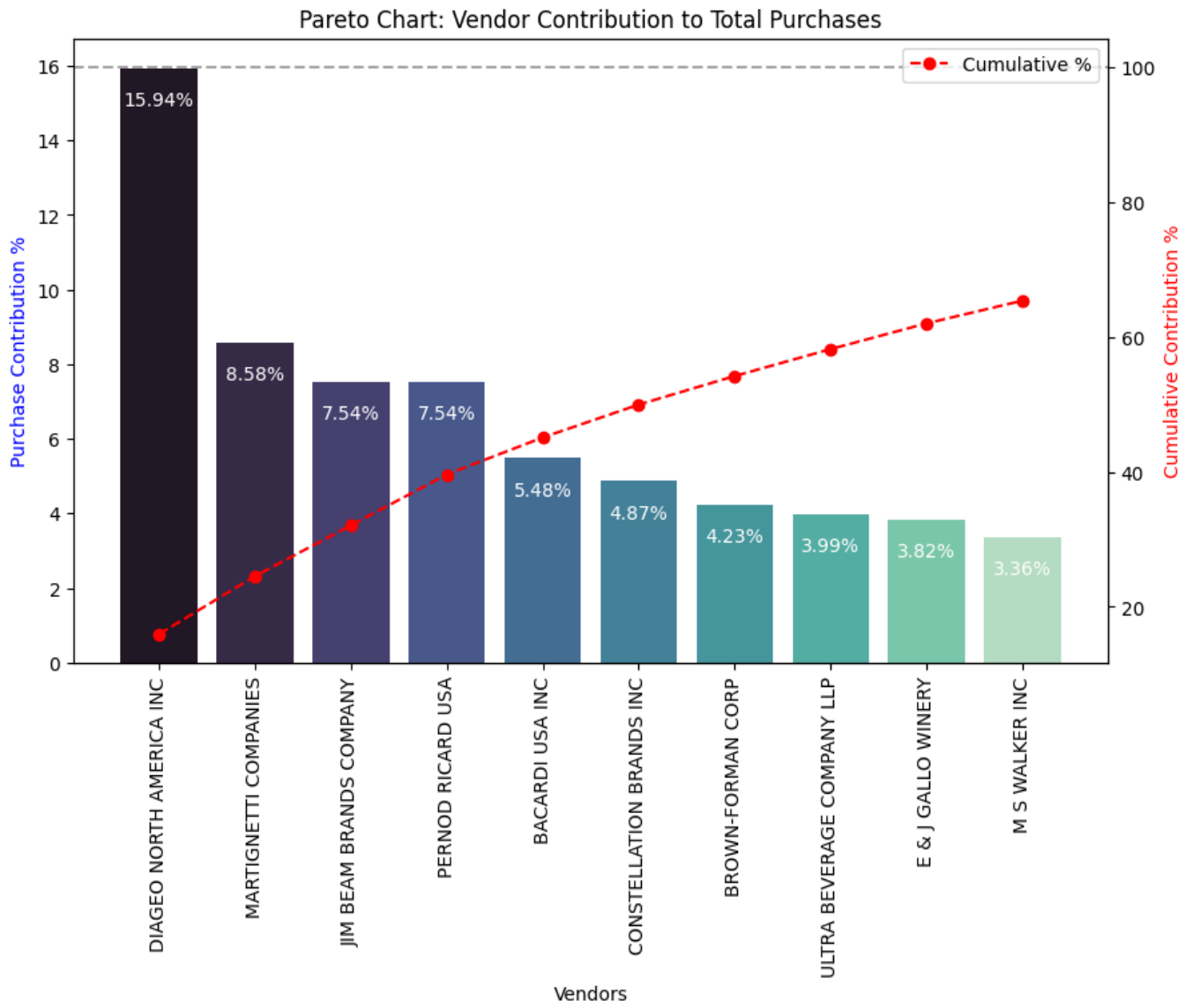




5.3 Vendor Concentration & Dependency

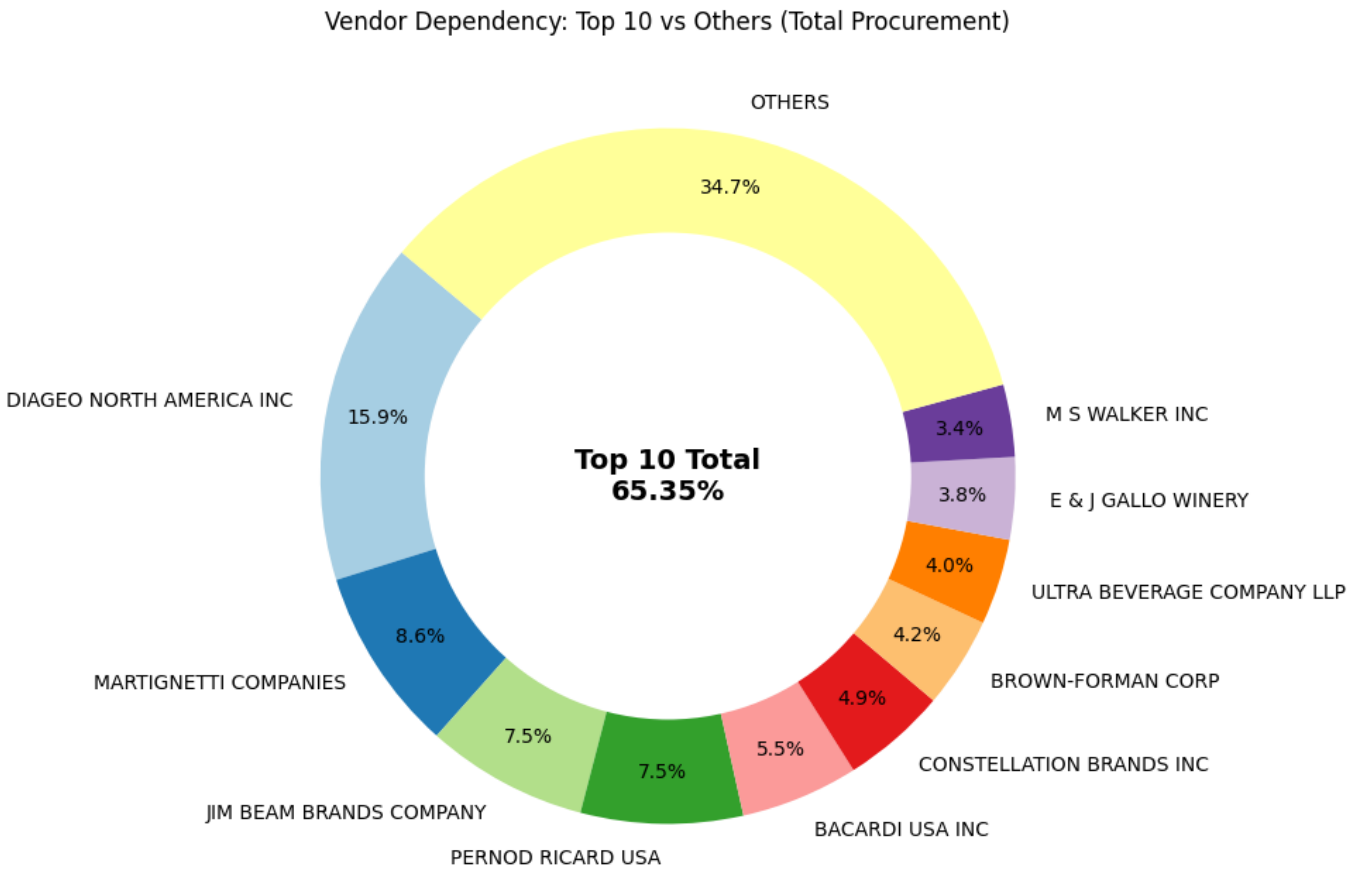
A **Pareto Analysis** (80/20 rule) was conducted to evaluate the supply chain's stability.

- **The "Vital Few":** The Top 10 vendors account for **65.35%** of total procurement value.
- **Primary Partner: DIAGEO NORTH AMERICA INC** dominates the portfolio, representing nearly **16%** of all purchases.
- **Risk Note:** While this concentration maximizes "Economies of Scale" (bulk discounts), it also increases the business's vulnerability to price changes or supply disruptions from these top partners.



The **Vendor Dependency Donut Chart** provides a strategic high-level view of the procurement landscape, complementing the Pareto Analysis.

- **Dominant Market Share**: The "Top 10" vendors collectively control **65.35%** of the total procurement volume. This confirms a high degree of market concentration where a small group of suppliers holds the majority of the business's inventory value.
- **The Fragmented Tail**: The "OTHERS" category accounts for **34.7%** of the total procurement. While this represents a large portion of the business, it is spread across a wider variety of smaller vendors, indicating a fragmented tail beyond the primary power players.



5.4 Inventory Velocity & Capital Health

To optimize cash flow, we identified vendors associated with **stagnant stock**—products that take the longest to sell.

- **Low Turnover Risks:** Vendors like **TRUETT HURST** (0.04) and **IRA GOLDMAN** (0.07) show critically low inventory velocity.

- **Unsold Value: DIAGEO** and **MARTIGNETTI** hold the highest dollar value of unsold inventory ($980K and $872K respectively). While they are top sellers, their sheer volume requires constant monitoring to ensure stock does not become dead capital.

VendorName	StockTurnover
TRUETT HURST	0.041667
IRA GOLDMAN AND WILLIAMS, LLP	0.075000
UNCORKED	0.217238
VINEYARD BRANDS LLC	0.256241
LOYAL DOG WINERY	0.308333
HIGHLAND WINE MERCHANTS LLC	0.335069
BLACK COVE BEVERAGES	0.340588
Russian Standard Vodka	0.394094
Serralles Usa LLC	0.414137
GILMANTON WINERY & VINEYARD	0.465819

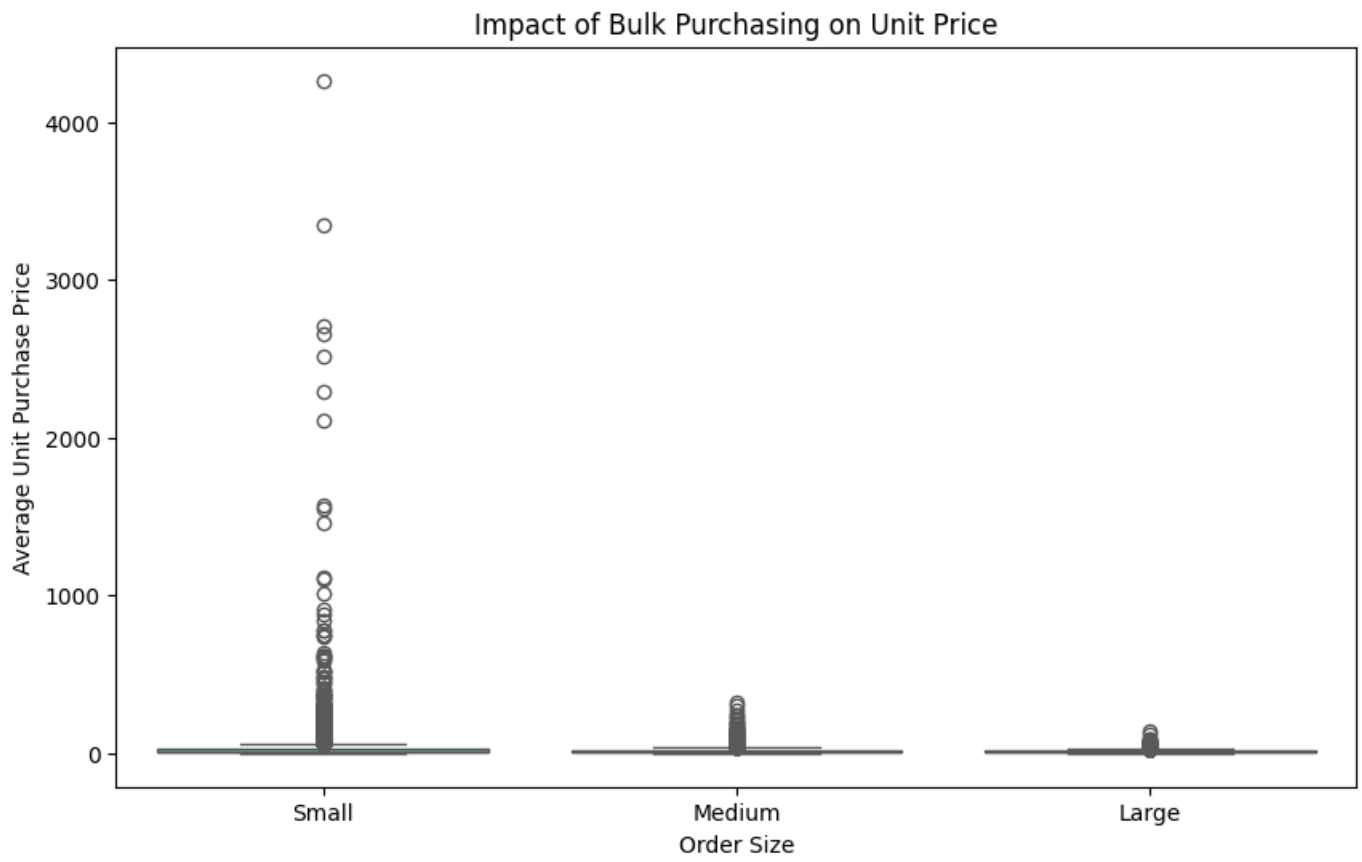
VendorName	UnsoldInventoryValue
DIAGEO NORTH AMERICA INC	980.08K
MARTIGNETTI COMPANIES	872.08K
JIM BEAM BRANDS COMPANY	810.30K
ULTRA BEVERAGE COMPANY LLP	715.98K
PERFECTA WINES	560.31K
PERNOD RICARD USA	554.74K
WILLIAM GRANT & SONS INC	436.49K
M S WALKER INC	397.54K
E & J GALLO WINERY	290.50K
BROWN-FORMAN CORP	284.91K

5.5 Procurement Optimization: The Power of Bulk

This section proves that "Buying Big" is the primary driver of cost efficiency.

The "73% Price Drop" (Box Plot Analysis) By segmenting orders into **Small, Medium, and Large**, we see a massive shift in cost:

- **Small Orders:** Highly expensive and volatile. Prices often spike into the thousands (outliers).
- **Large Orders:** Prices stabilize at their lowest point. Moving from Small to Large saves approximately **73%** per unit.
- **Significant Inverse Relationship:** There is a clear, strong inverse relationship between order size and purchase price; as the quantity increases, the cost per unit drops drastically.
- **Action:** Eliminate small, fragmented orders to stop margin erosion.



6. Statistical Validation: Hypothesis Testing & Confidence Intervals

This section uses inferential statistics to move beyond simple averages and confirm the reliability of our business insights with **95% confidence**.

6.1 Confidence Interval Comparison: Top vs. Low Vendors

To ensure our findings aren't just a result of random chance, we performed a **Confidence Interval (CI) Comparison** between the profit margins of high-performing and low-performing vendors.

The Methodology:

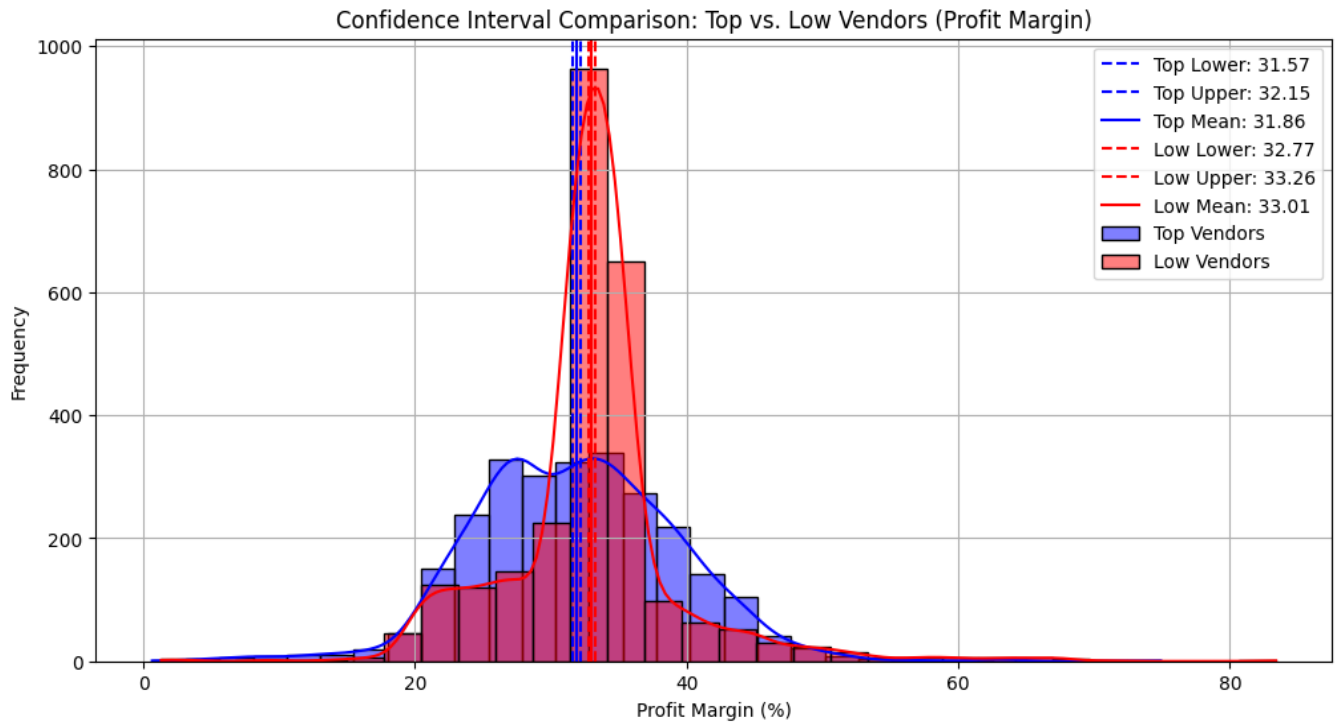
- **Top Vendors:** The top 25% of the fleet based on TotalSalesDollars.
- **Low Vendors:** The bottom 25% of the fleet based on TotalSalesDollars.
- **The Goal:** Determine if "Top Vendors" actually yield higher profit margins than "Low Vendors" with 95% statistical certainty.

Key Findings:

- **Top Vendors:** 95% CI: **(31.57%, 32.15%)** | Mean: **31.86%**
- **Low Vendors:** 95% CI: **(32.77%, 33.26%)** | Mean: **33.01%**

Statistical Insight: Interestingly, the **Low-Volume vendors actually maintain a higher average profit margin** (~1.15% higher) than the high-volume giants. Since the confidence intervals do not overlap, this difference is **statistically significant**.

Business Implication: This confirms the "High-Volume, Low-Margin" nature of top players like **DIAGEO**. While they drive the bulk of the revenue, their margins are compressed due to competitive pricing and bulk trading. Conversely, the smaller vendors in the "Fragmented Tail" offer higher margins per unit, validating our previous strategy to target these "Hidden Gems" for growth.



6.2 Statistical Testing: Independent T-Test Analysis

To further solidify our confidence interval findings, we conducted a formal **Independent Two-Sample T-Test** to compare the mean profit margins of top-performing and low-performing vendors.

- **Null Hypothesis (H0):** There is no significant difference in the mean profit margins between the two groups.
- **Alternative Hypothesis (H1):** There is a statistically significant difference in their mean profit margins.

Test Results

- **T-Statistic:** -5.9978
- **P-Value:** 0.0000 (Highly Significant)

The Verdict: Reject H0

Since the P-Value is far below the standard alpha threshold of 0.05, we **reject the Null Hypothesis**. This provides absolute statistical proof that the difference in profit margins between these two vendor groups is not due to random noise in the data.

Business Interpretation:

The negative T-Statistic confirms the direction of the difference: **Low-performing (smaller) vendors have a significantly higher mean profit margin than top-performing vendors.** This validates the strategic pivot mentioned in Section 5.1 while the Top 10 vendors provide the necessary volume for cash flow, the business must look toward smaller, niche vendors to drive higher profitability per unit.

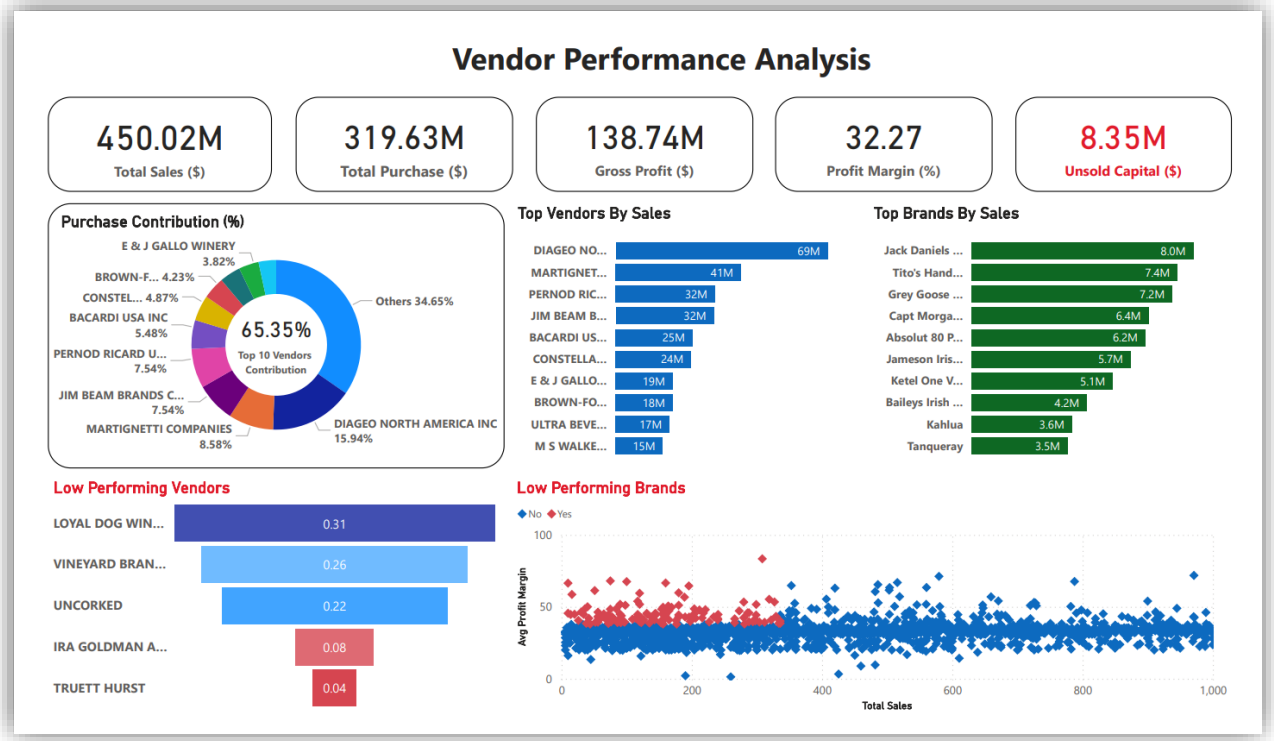
7. Vendor Performance Dashboard

The Vendor Performance Power BI Dashboard provides a consolidated view of financial performance, vendor contribution, and profitability. Key metrics show strong business performance with **$450.02M in total sales**, **$138.74M gross profit**, and a healthy **32.27% profit margin**, while **$8.35M unsold capital** highlights inventory optimization needs.

A Pareto analysis reveals that the **Top 10 vendors contribute 65.35%** of total procurement, indicating high dependency on a few key suppliers. Top vendors like DIAGEO and leading brands such as Jack Daniels drive the majority of revenue, ensuring stable cash flow. However, low-performing vendors exhibit poor inventory turnover, contributing to stagnant stock and tied-up capital.

The sales vs. profit margin analysis highlights a key trade-off: **high-volume vendors generate consistent revenue with lower margins, while smaller vendors offer higher profitability per unit**.

Overall Insight: The dashboard emphasizes the need for a balanced procurement strategy maintaining strong relationships with top vendors for volume while expanding high-margin, low-volume vendors to improve profitability and reduce risk.



8. Conclusion

The project demonstrates that while high-volume market leaders like **DIAGEO** provide the necessary cash flow, the true "margin drivers" are found within the fragmented tail of smaller vendors. The data-driven results lead to two primary strategic mandates:

1. **Mandatory Order Consolidation:** To stop margin erosion, the procurement strategy must shift away from small, fragmented orders. Moving into the "Large" volume tier is the most direct lever for achieving the identified **73% cost savings**.

2. **Strategic Portfolio Balancing:** The business should protect its core volume with Top 10 vendors while aggressively expanding the "Hidden Gems" segment brands with margins above 38% to improve overall capital utilization.

By integrating **advanced SQL** aggregation, **Python** with **statistics**, this pipeline provides a robust framework for continuous inventory optimization and sustainable profit growth.